

Explainable Vision-Language Diagnostic Assistant for Chest Radiographs: An Offline and Edge-Optimized Multimodal Framework

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Abstract—Rural and resource-constrained healthcare centres may face limited access to specialist radiologists, resulting in delays in the interpretation of chest radiographs. Existing medical artificial intelligence systems can provide disease classification, but classification probabilities alone may be difficult for non-specialist healthcare workers to interpret and trust. In addition, cloud-based medical AI systems may be unsuitable for locations with unreliable internet connectivity.

This paper proposes an Explainable Vision-Language Diagnostic Assistant for Chest Radiographs that combines automated disease classification, visual explanation, natural-language generation, conservative triage support, and offline edge deployment. The proposed framework uses a Vision Transformer (ViT)-based vision encoder for spatial feature extraction, CLIP-style or medical contrastive alignment for image-text representation, and a BLIP-2-inspired generation component for producing concise plain-language diagnostic descriptions. Grad-CAM heatmaps and attention visualization are incorporated to provide visual evidence associated with model predictions. A separate rule-based triage layer maps validated model outputs to conservative priority categories.

The system is designed for offline execution on a mid-range Android device using model optimization, INT8 quantization, and ONNX/mobile runtime conversion. The evaluation framework considers disease classification performance, explanation quality, text-generation quality, triage sensitivity, robustness, and mobile runtime performance. The proposed system is intended as a clinical decision-support and screening assistant rather than an autonomous diagnostic system.

Index Terms—Chest Radiographs, Medical AI, Vision-Language Models, Explainable AI, Vision Transformer, CLIP, BLIP-2, Grad-CAM, Medical Image Classification, Edge AI, Offline AI, Clinical Decision Support

I. INTRODUCTION

A. Background

Chest radiography is an important imaging modality for identifying thoracic abnormalities and supporting clinical decision-making. However, interpretation of radiographs often requires specialist knowledge and experience. In rural and resource-constrained healthcare settings, access to radiologists may be limited, which can contribute to diagnostic delays and increased travel requirements for patients.

Recent advances in deep learning and medical computer vision have enabled automated analysis of chest radiographs.

Deep convolutional neural networks and transformer-based architectures can learn representations from medical images and predict multiple thoracic conditions. However, a classification result alone does not necessarily explain why a particular prediction was generated.

Explainability is therefore important when developing AI-based medical decision-support systems. Visual attribution methods such as Grad-CAM can highlight image regions contributing to a prediction, while vision-language models can provide textual descriptions that are more accessible to non-specialist users.

The proposed project combines these capabilities into a unified multimodal framework. The system accepts a chest radiograph, performs image preprocessing and disease prediction, generates visual explanations, produces a plain-language diagnostic description, and provides a conservative priority indicator.

The proposed system is specifically designed with offline deployment in mind. Model optimization and INT8 quantization are considered to enable execution on an Android or equivalent edge device without continuous internet connectivity.

B. Problem Statement

The major challenges addressed by this research are:

- 1) Limited access to specialist radiologists in some rural healthcare settings.
- 2) Diagnostic delays and increased travel burden for patients.
- 3) Lack of transparency in classification-only AI systems.
- 4) Difficulty for non-specialist healthcare workers to interpret raw model probabilities.
- 5) Dependence of cloud-based AI systems on continuous internet connectivity.
- 6) Computational limitations of deploying large vision-language models on mobile and edge devices.

The proposed research addresses these challenges through an explainable multimodal architecture that integrates image classification, visual attribution, language generation, conservative triage support, and offline deployment.

C. Research Gap

The literature reviewed for this project demonstrates substantial progress in medical image classification, explainable AI, and vision-language learning. However, several components are frequently addressed independently.

Classification architectures such as DenseNet and CheXNet focus primarily on disease prediction. Grad-CAM provides visual attribution but does not itself generate natural-language diagnostic explanations. Vision-language approaches such as BLIP-2, MedKLIP, GLoRIA, ConVIRT, and MedCLIP improve image-text representation or generation but can remain computationally demanding or may not provide an integrated offline diagnostic workflow.

The research gap identified for this project is therefore the lack of an integrated framework that combines:

- multi-label chest-radiograph classification,
- visual explanation,
- vision-language representation,
- plain-language diagnostic generation,
- conservative triage support, and
- offline edge deployment.

D. Research Objectives

The objectives of the proposed research are:

- 1) To develop a multimodal assistant for chest-radiograph analysis.
- 2) To generate disease predictions with calibrated confidence scores.
- 3) To generate Grad-CAM heatmaps and attention-based explanations.
- 4) To generate concise plain-language diagnostic descriptions.
- 5) To develop a conservative rule-based triage-support layer.
- 6) To optimize the model for offline edge execution.
- 7) To evaluate classification, explanation, language generation, triage, robustness, and runtime performance.

E. Research Questions

The study investigates the following research questions:

- RQ1: How effectively can a multimodal architecture classify thoracic abnormalities from chest radiographs?
- RQ2: Can Grad-CAM and attention visualization provide useful visual evidence for model predictions?
- RQ3: Can image-text alignment improve the generation of concise diagnostic descriptions?
- RQ4: Can a conservative rule-based layer provide useful priority guidance without presenting itself as a definitive diagnosis?
- RQ5: What effect does INT8 quantization have on model performance and edge-device runtime?
- RQ6: Can the complete system operate without continuous internet connectivity?

F. Contributions

The main contributions of this research are:

- 1) A modular vision-language architecture for chest-radiograph decision support.
- 2) Integration of image classification and language generation.
- 3) Integration of Grad-CAM and attention-based visual explanations.
- 4) A separated conservative triage-support module.
- 5) An offline edge-deployment strategy using model optimization and INT8 quantization.
- 6) A multi-dimensional evaluation framework covering prediction, explanation, language, safety, robustness, and runtime performance.

II. RELATED WORK

A. Deep Learning for Medical Image Classification

Deep learning architectures have demonstrated strong performance in image recognition. ResNet introduced residual learning through shortcut connections, enabling the construction of deeper neural networks [1]. DenseNet subsequently introduced dense connections between layers to improve feature propagation and feature reuse [2].

These architectures provide important foundations for medical image classification but were not originally designed as complete vision-language diagnostic systems.

B. Chest X-Ray Disease Classification

CheXNet demonstrated the application of a 121-layer DenseNet to pneumonia detection in chest radiographs and extended classification to multiple diseases in the ChestX-ray14 dataset [3]. Such systems demonstrate the feasibility of automated chest X-ray classification.

However, classification-oriented approaches generally do not provide a complete workflow consisting of natural-language explanation, visual evidence, conservative triage, and offline mobile deployment.

C. Explainable Artificial Intelligence

Grad-CAM introduced a gradient-based technique for producing class-discriminative localization maps that identify image regions important for a model prediction [4].

For medical applications, such visual attribution can provide useful evidence regarding which regions influenced a prediction. Nevertheless, a heatmap should be interpreted as model-attribution evidence rather than definitive proof of pathology.

D. Vision-Language Models

BLIP-2 demonstrated an approach for connecting frozen image encoders with large language models using a lightweight Querying Transformer (Q-Former) [5]. Its architecture provides motivation for using a separate visual representation and language-generation component.

Medical vision-language methods have also explored the alignment between medical images and textual reports. Med-KLIP incorporates medical knowledge and spatial alignment into radiology image-language pre-training [6]. GLoRIA learns global and local image-text representations by associating image regions with report words [7]. ConVIRT applies contrastive learning to paired medical images and text [8], while MedCLIP explores contrastive learning from medical images and text using medical knowledge [9].

E. Research Gap from Existing Approaches

Table I summarizes the relationship between representative approaches and the proposed system.

III. PROPOSED METHODOLOGY

A. Overall Framework

The proposed methodology follows a modular pipeline that transforms a chest radiograph into disease predictions, visual explanations, plain-language descriptions, and conservative priority guidance.

The overall workflow is:

$$\text{X-ray} \rightarrow \text{Preprocessing} \rightarrow \text{Vision Encoder} \rightarrow \text{Classification} \quad (1)$$

$$\rightarrow \text{Explainability} \rightarrow \text{Vision-Language Generation} \rightarrow \text{Triage} \rightarrow \text{Offline Application.} \quad (2)$$

B. Input Layer

The system accepts chest radiographs in formats such as PNG, JPEG, or DICOM after conversion to the required model input format.

The input stage performs:

- image format validation,
- quality checks,
- resizing,
- normalization, and
- training-time augmentation.

C. Preprocessing

Let the input image be represented as:

$$I \in \mathbb{R}^{H \times W \times C}.$$

The preprocessing function can be represented as:

$$I' = \mathcal{P}(I)$$

where $\mathcal{P}$ represents resizing, normalization, and quality processing.

Training augmentation is applied only to the training data in order to reduce overfitting while preventing information leakage.

D. Vision Encoder

A Vision Transformer (ViT)-based encoder is proposed for extracting spatial image representations.

The image is divided into patches and converted into a sequence of embeddings. The encoder produces a feature representation:

$$V = f_{\theta}(I')$$

where f_{θ} represents the vision encoder and V represents the extracted visual features.

E. Disease Prediction Head

The disease prediction head performs multi-label thoracic disease classification.

For K disease categories, the prediction vector is:

$$\hat{y} = [\hat{y}_1, \hat{y}_2, \dots, \hat{y}_K]$$

where each $\hat{y}_k$ represents the estimated probability of the corresponding abnormality.

A sigmoid activation can be used:

$$\hat{y}_k = \frac{1}{1 + e^{-z_k}}$$

where z_k is the classification logit for disease k .

The system reports both predicted abnormalities and associated confidence values.

F. Vision-Text Alignment

The vision-language component maps image and textual representations into a shared embedding space.

Let:

$$V = f_{\theta}(I)$$

represent the image embedding and

$$T = g_{\phi}(R)$$

represent the text embedding.

The similarity between image and text representations can be expressed as:

$$S(V, T) = \frac{V \cdot T}{\|V\| \|T\|}$$

where $S(V, T)$ represents cosine similarity.

This alignment provides a mechanism for connecting radiographic visual features with medically relevant textual concepts.

TABLE I
COMPARISON OF REPRESENTATIVE EXISTING APPROACHES

Work	Main Approach	Major Contribution	Limitation Relevant to This Study
ResNet [1]	Residual CNN	Deep image representation	Not inherently designed for multimodal radiology, language generation, or triage
DenseNet [2]	Dense CNN	Feature reuse and propagation	Primarily image classification architecture
CheXNet [3]	DenseNet-based classifier	Chest X-ray disease classification	Does not provide complete vision-language explanation and offline triage workflow
Grad-CAM [4]	Gradient-based attribution	Visual localization	Does not generate textual diagnostic explanations
BLIP-2 [5]	Vision-language model	Image-to-language generation	Computational requirements may challenge small edge devices
MedKLIP [6]	Medical vision-language learning	Medical knowledge and spatial alignment	Does not directly address offline mobile deployment and triage
GLoRIA [7]	Global-local image-text learning	Region-text representation learning	Requires paired image-report data and does not provide the complete proposed workflow
ConVIRT [8]	Contrastive learning	Medical image-text representation	Primarily focuses on transferable representations
MedCLIP [9]	Medical contrastive learning	Image-text semantic alignment	Not a complete diagnostic-to-triage assistant
Proposed System	Multimodal + Explainable + Edge	Classification, explanation, generation and triage	Designed specifically for offline edge-oriented chest-radiograph decision support

G. Language Generation

A BLIP-2-inspired or lightweight image-to-text generation component is used to produce concise plain-language descriptions.

The generated output can be represented as:

$$R = G(V, Y)$$

where V represents visual features and Y represents relevant prediction information.

The generation component is constrained to reduce unsupported findings and hallucinated diagnoses. Where possible, generated descriptions are aligned with detected findings and predefined output structures.

H. Explainability Layer

The proposed system uses Grad-CAM to generate class-specific heatmaps.

For a target class c , Grad-CAM computes importance weights from gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

where A^k is an activation map and y^c is the score for class c .

The resulting localization map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right).$$

The resulting heatmap highlights regions that contributed to the model's prediction.

Cross-attention visualization may also be included where supported by the selected vision-language architecture.

I. Triage Support Layer

The system includes a separate rule-based triage-support layer.

Rather than presenting triage as a diagnosis, validated prediction outputs are mapped to conservative categories such as:

- Routine review,
- Priority review,
- Urgent specialist review.

The triage layer is separated from the disease prediction model so that the safety rules can be independently inspected and validated.

J. Edge Optimization

The final model is optimized for offline execution.

The optimization pipeline includes:

- 1) Model profiling,
- 2) Freezing large pretrained components where appropriate,
- 3) Precision reduction,
- 4) FP16 evaluation,
- 5) INT8 quantization,
- 6) ONNX conversion where compatible,
- 7) Mobile runtime integration.

The effect of optimization is evaluated in terms of model size, memory usage, latency, and predictive performance.

K. End-to-End System Flow

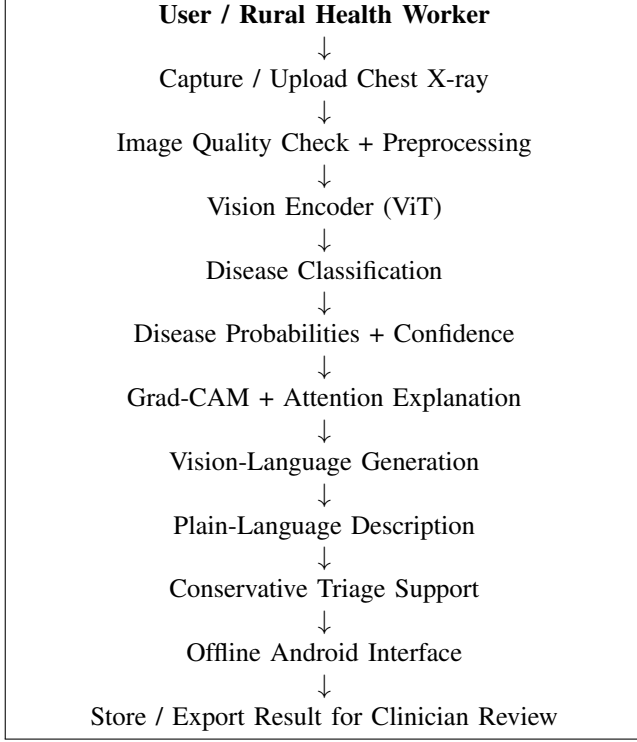


Fig. 1. End-to-end workflow of the proposed diagnostic assistant.

IV. DATASETS AND EXPERIMENTAL SETUP

A. Datasets

The project considers publicly available chest-radiograph datasets including ChestX-ray14, OpenI, and ROCO. CheXpert and MIMIC-CXR may also be considered where access and project scope permit.

TABLE II
DATASETS CONSIDERED FOR THE PROPOSED STUDY

Dataset	Primary Use	Data Type
ChestX-ray14	Multi-label classification	Chest radiographs + disease labels
OpenI	Image-report learning	Radiographs + reports
ROCO	Image-text learning	Medical images + captions
CheXpert	Classification / validation	Radiographs + reports/labels
MIMIC-CXR	Image-report research	Radiographs + radiology reports

The project emphasizes patient-wise data splitting wherever patient identifiers are available in order to reduce data leakage between training and testing subsets.

B. Data Preprocessing

The preprocessing pipeline consists of:

- 1) Removal of corrupt or duplicate images.
- 2) Standardization of image formats.

- 3) Patient-wise train, validation, and test splitting.
- 4) Image resizing.
- 5) Pixel normalization.
- 6) Training-only augmentation.
- 7) Multi-label target preparation.
- 8) Preparation of image-report pairs where applicable.

C. Baseline Models

A baseline classifier based on DenseNet or a CheXNet-style architecture is considered for comparison with the proposed multimodal framework.

The baseline establishes a reference point for measuring the impact of the additional vision-language and explainability components.

D. Implementation Technologies

The proposed implementation may use:

- Python,
- PyTorch,
- Hugging Face Transformers,
- OpenCV,
- NumPy,
- ONNX Runtime,
- Android/mobile inference runtime,
- Grad-CAM implementation libraries.

V. EVALUATION METHODOLOGY

The proposed system is evaluated across multiple dimensions rather than using classification accuracy alone.

A. Disease Classification

The classification component is evaluated using:

- Precision,
- Recall,
- F1-score,
- ROC-AUC,
- Sensitivity,
- Specificity.

Precision is defined as:

$$Precision = \frac{TP}{TP + FP}$$

and recall is:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score is:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

B. Visual Explanation Evaluation

Explanation quality is evaluated using localization checks when ground-truth region annotations are available.

Where appropriate, the evaluation may include:

- localization overlap,
- pointing-game style evaluation,
- qualitative expert review,
- comparison between predicted regions and annotated regions.

The heatmaps are interpreted as model-attribution evidence rather than as direct proof of disease.

C. Text Generation Evaluation

Generated descriptions are evaluated using:

- Clinical factuality review,
- Standard text-similarity metrics where appropriate,
- Unsupported-finding analysis,
- Hallucination analysis.

The primary safety consideration is that the generated description should not introduce findings unsupported by the image or validated model output.

D. Triage Evaluation

The triage-support component is evaluated using:

- Sensitivity for high-priority cases,
- False-negative analysis,
- Rule consistency,
- Safety-oriented error analysis.

The triage output is presented as decision support and not as a definitive clinical diagnosis.

E. Edge Deployment Evaluation

The mobile/edge implementation is evaluated using:

- Inference latency,
- Peak RAM usage,
- Model size,
- Runtime behavior,
- Battery impact where measurable.

FP32, FP16, and INT8 versions are compared to determine the effect of quantization.

F. Ablation Studies

The following ablation studies are planned:

- 1) Classifier only vs. classifier + Grad-CAM.
- 2) Image-only representation vs. image-text alignment.
- 3) Full precision vs. INT8 inference.
- 4) Classification-only output vs. multimodal output.
- 5) Triage with and without conservative safety rules.

VI. RESULTS

Note: This section should be updated with measured experimental results after model training and evaluation. No numerical performance values are claimed here before experiments are completed.

A. Classification Results

Table III can be used to report the classification results.

TABLE III
DISEASE CLASSIFICATION PERFORMANCE

Model	Precision	Recall	F1	ROC-AUC
DenseNet Baseline	–	–	–	–
Proposed Vision Encoder	–	–	–	–
Proposed Multimodal Model	–	–	–	–

B. Explanation Results

The explanation experiments will compare generated Grad-CAM and attention maps with available localization annotations and/or expert qualitative assessment.

TABLE IV
VISUAL EXPLANATION EVALUATION

Evaluation Method	Result
Localization Overlap	–
Pointing-Game Evaluation	–
Expert Qualitative Review	–
Failure Cases	–

C. Text Generation Results

The generated diagnostic descriptions will be evaluated for factual consistency and language quality.

TABLE V
TEXT GENERATION EVALUATION

Metric	Result
BLEU	–
ROUGE	–
Clinical Factuality	–
Unsupported Findings	–
Hallucination Rate	–

D. Edge Deployment Results

TABLE VI
EDGE DEPLOYMENT PERFORMANCE

Model	Size	Latency	RAM
FP32	–	–	–
FP16	–	–	–
INT8	–	–	–

VII. DISCUSSION

The proposed framework is designed to address limitations associated with isolated medical AI components.

A classification-only system can provide disease probabilities but may not provide sufficient visual or textual context for interpretation. The addition of Grad-CAM provides a visual representation of regions associated with a prediction. The vision-language component provides a mechanism for

converting visual representations and detected findings into a concise textual description.

The separation of the triage-support layer from the disease prediction model is an important design consideration. Rather than directly converting a model probability into a clinical diagnosis, the system uses validated thresholds and conservative safety rules to generate a priority-support indication.

Offline deployment is another important aspect of the proposed framework. Cloud-based inference may be difficult in environments with unreliable connectivity. Model compression and INT8 quantization can reduce computational requirements, although quantization may introduce some loss of predictive performance. Therefore, the optimized model must be evaluated against the original model.

The framework also emphasizes uncertainty. A visual explanation does not prove the presence of pathology, and generated language should not be interpreted as independent clinical evidence. These limitations should be clearly communicated in the final user interface.

VIII. PRACTICAL IMPLICATIONS

The proposed system is intended primarily as a decision-support tool for resource-constrained healthcare environments.

Potential applications include:

- Preliminary chest-radiograph screening.
- Support for rural health workers.
- Visual explanation of model predictions.
- Generation of simplified diagnostic descriptions.
- Conservative prioritization of cases for specialist review.
- Offline medical-AI experimentation and demonstration.

The system should not replace radiologists or other qualified healthcare professionals. Medical decisions should remain subject to clinical confirmation.

IX. LIMITATIONS

Several limitations are anticipated.

- 1) **Dataset label noise:** Some chest-radiograph labels may be derived automatically from reports and may contain inaccuracies.
- 2) **Distribution shift:** Performance may change when the model is applied to images from hospitals, devices, populations, or acquisition protocols that differ from the training data.
- 3) **Generated-text hallucination:** A vision-language model may generate unsupported findings. Therefore, generation must be constrained and evaluated for factual consistency.
- 4) **Explanation faithfulness:** Grad-CAM and attention maps represent model attribution and should not automatically be interpreted as clinically verified pathology localization.
- 5) **Mobile computation:** Large vision-language models may be computationally expensive for mobile devices.
- 6) **Triage safety:** The proposed triage mechanism is a decision-support component and cannot independently establish clinical urgency.

- 7) **Dataset availability:** Some datasets, such as MIMIC-CXR and CheXpert, may require controlled access or additional approvals.

X. CONCLUSION AND FUTURE WORK

This research proposes an Explainable Vision-Language Diagnostic Assistant for Chest Radiographs that integrates medical image classification, visual explanation, language generation, conservative triage support, and offline edge deployment.

The proposed architecture combines a ViT-based vision encoder, CLIP-style or medical contrastive image-text alignment, and a BLIP-2-inspired generation component. Grad-CAM and attention visualization provide visual evidence associated with model predictions, while a separate rule-based layer provides conservative priority guidance.

A major focus of the framework is offline operation. Model optimization, INT8 quantization, and ONNX/mobile runtime conversion are proposed to reduce computational requirements and enable deployment on resource-constrained devices.

The system is intended as clinical decision support rather than an autonomous diagnostic device. The evaluation framework therefore includes not only disease classification performance but also explanation quality, language factuality, triage sensitivity, robustness, and mobile runtime behavior.

Future work will focus on:

- 1) External validation on additional chest-radiograph datasets.
- 2) More extensive patient-wise and cross-dataset evaluation.
- 3) Improved factual consistency for generated descriptions.
- 4) More rigorous evaluation of explanation faithfulness.
- 5) Improved medical vision-language alignment.
- 6) Further model compression and mobile optimization.
- 7) Integration with additional Android-supported inference runtimes.
- 8) Expert evaluation involving qualified radiologists.
- 9) Investigation of multilingual diagnostic descriptions.
- 10) More comprehensive safety and failure-case analysis.

DECLARATIONS

Funding

No external funding is associated with this academic project.

Conflict of Interest

The authors declare that there is no conflict of interest related to this research project.

Ethics Statement

The proposed research is based on publicly available or appropriately authorized medical imaging datasets. The system is intended for research and decision-support purposes and not for autonomous clinical diagnosis.

Data Availability

The study considers publicly accessible medical imaging datasets such as ChestX-ray14, OpenI, and ROCO. Additional datasets such as CheXpert and MIMIC-CXR may be used subject to their respective access requirements.

Author Contributions

P. Kavya Sai: Project architecture, research design, implementation, documentation, and evaluation.

P. Meghna: Dataset preparation, model development, and experimental analysis.

S. Anjana: Vision-language components, explainability, and evaluation.

D. Poojitha: Application development, testing, documentation, and presentation.

All authors contributed to the development and review of the project.

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